

Rossmann Retail Demand Forecasting & Promotion Analysis

Key Project Insights

This document summarizes the analytical, statistical, and modeling insights from the exploratory data analysis, classical time series forecasting, machine learning forecasting, and promotion analysis.

1. Exploratory Data Analysis (EDA) & Demand Dynamics

Weekly Seasonality

- Retail sales show a strong 7-day cyclical pattern.
- Monday experiences the highest weekly demand.
- Standard Sunday closures cause aggregate sales to drop to near-zero.

Monthly & Annual Patterns

- Significant seasonal sales spikes occur in December, coinciding with the Christmas and holiday shopping period.
- Sales typically decline after the holiday period in January.
- End-of-month and mid-month changes in sales were also observed.

Store Status

- Approximately 17% of historical records correspond to closed stores, including Sundays, state holidays, and refurbishments.
- Closed-store observations have zero sales and can distort baseline promotion comparisons.
- For promotion analysis, open-store observations (Open == 1) were used to avoid including closed-store zero-sales observations.

Store Typology

- Store Type B generates the highest average daily sales at approximately **10,058**.
- Store Types A, C, and D have similar average sales of approximately **5,600–5,740**.
- Store Type D shows higher sales per customer.

Assortment

Assortment Type	Mean Sales
Extra	~8,554

Assortment Type	Mean Sales
Extended	~6,059
Basic	~5,481

Stores offering Extra assortment have higher mean sales compared with Basic and Extended assortments.

Customers and Sales

- Customer count has the strongest contemporaneous relationship with sales, with a Pearson correlation of approximately **0.89**.
 - Promotional periods are associated with approximately **58.3%** higher customer footfall.
 - Average spend per customer increased by approximately **13.9%** during promotional periods, from **8.94** to **10.18**.
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2. Classical Time Series Forecasting

Evaluation Framework

- All forecasting models were evaluated on the same 60-day chronological test period.
- **Test period:** 2015-06-02 to 2015-07-31
- The chronological holdout was used to avoid using future observations when evaluating forecasting performance.

Naïve Model

- **Model:** Naïve / Last Value
- **MAE:** 2,746,029
- **RMSE:** 4,278,452
- The model struggled to anticipate weekly demand changes and produced large forecast errors.

Seasonal Naïve

- **Model:** Seasonal Naïve using a 7-day lag
- **MAE:** 2,084,718
- **RMSE:** 2,570,396
- Captured the strong weekly sales pattern.
- Improved MAE by approximately **24.1%** compared with the standard Naïve model.

ARIMA

- **Model:** ARIMA(7,0,0)

- **MAE:** 2,075,115
- **RMSE:** 2,727,432
- Improved slightly over Seasonal Naive on MAE.
- Captured short-term autoregressive patterns but struggled with some holiday and calendar changes.

SARIMA

- **Model:** SARIMA(1,0,0)x(1,0,1,7)
- **MAE:** 1,122,513
- **RMSE:** 1,540,606
- Best-performing classical forecasting model.
- Reduced MAE by approximately **45.9%** compared with ARIMA.
- Explicitly modeled weekly seasonal structure.

Classical Model Limitations

- The ARIMA and SARIMA models used in this project relied primarily on historical sales and seasonal structure.
 - They did not directly use external variables such as promotions or holidays.
 - This limits their ability to model changes associated with external factors.
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3. Machine Learning Forecasting — LightGBM

Model Performance

LightGBM was the best-performing model among the five evaluated forecasting approaches.

- **MAE:** 323,834
- **RMSE:** 451,189

Compared with the best classical model, SARIMA:

- **MAE reduction:** 71.15%
- **RMSE reduction:** 70.71%

Feature Engineering

The LightGBM model used temporal, lag, rolling-window, calendar, and promotion-related features.

Lag Features

- lag_7
- lag_14
- lag_28

Rolling Features

- rolling_mean_7
- rolling_mean_14
- rolling_std_7

Calendar and External Features

- Day of week
- Day of month
- Month
- Promotion status
- Holiday indicators

Top Model Features

1. lag_7
2. rolling_mean_7
3. Promo
4. DayOfWeek
5. lag_14

The feature importance results indicate that recent historical sales, weekly patterns, and promotion status were among the most useful features for the LightGBM model.

4. Forecasting Model Performance Summary

Rank	Model	MAE	RMSE
1	LightGBM	323,834	451,189
2	SARIMA(1,0,0)x(1,0,1,7)	1,122,513	1,540,606
3	ARIMA(7,0,0)	2,075,114	2,727,432
4	Seasonal Naive	2,084,718	2,570,396
5	Naive	2,746,029	4,278,452

MAE Rank

1. LightGBM
2. SARIMA
3. ARIMA

4. Seasonal Naive
5. Naive

RMSE Rank

1. LightGBM
2. SARIMA
3. Seasonal Naive
4. ARIMA
5. Naive

LightGBM improvement compared with SARIMA:

- MAE: 71.15% lower
 - RMSE: 70.71% lower
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5. Promotion Analysis

Open-Store Baseline Comparison

Analysis population: 844,392 open-store observations

Group	Observations	Percentage	Mean Sales	Median Sales
No Promotion	467,496	55.36%	5,929	5,459
Promotion	376,896	44.64%	8,228	7,649

Unadjusted comparison:

- Absolute difference: **+2,299 sales/day**
- Relative difference: **+38.77%**

Sales were substantially higher on promotion-active days in the open-store analysis.

Within-Store Comparison

- Total stores with both promotion and non-promotion observations: **1,115**
- Stores with higher average sales during promotions: **1,114**
- Stores with lower average sales during promotions: **1**
- Percentage of stores with higher promotional sales: **99.91%**

The positive association between promotions and sales was therefore observed across almost the entire store network.

Calendar Stratification

Sales were also compared within groups sharing:

- Day of week
- School holiday status
- State holiday status

Results:

- Calendar groups analyzed: **25**
- Groups with higher average sales during promotions: **24**
- Percentage: **96.00%**

OLS Regression

Model specification:

```
Sales ~ Promo + C(Store) + C(DayOfWeek) + SchoolHoliday + C(StateHoliday)
```

Analysis population: 844,392 open-store observations

Statistic	Value
Promotion coefficient	+2,293.66
Standard error	3.63
t-statistic	631.1
p-value	< 0.0001
95% CI (lower)	2,286.54
95% CI (upper)	2,300.79

Interpretation

After controlling for store identity, day of week, school holidays, and state holidays, promotion-active observations were associated with approximately **2,294 higher sales** on average.

The adjusted estimate of +2,293.66 was very close to the unadjusted open-store difference of +2,298.87. This indicates that the included store and calendar controls did not materially change the estimated promotion association.

However, this does not establish causality.

6. Limitations and Confounding

The promotion analysis is based on observational data.

The regression controls for:

- Store-level differences
- Day of week
- School holidays
- State holidays

However, other factors may influence both promotion timing and sales.

Potential unobserved factors include:

- Seasonal demand patterns
- Store-specific promotion strategies
- Local events
- Competitor activity
- Product availability
- Other business decisions

Promotions may also be scheduled during periods when higher sales are already expected.

Therefore, the regression coefficient should be interpreted as an **adjusted observational association rather than a proven causal effect**.

7. Strategic Business Applications

1. Demand Planning

- LightGBM substantially outperformed the evaluated classical forecasting models.
- The 71.15% MAE reduction and 70.71% RMSE reduction suggest that machine learning forecasting could be useful as an input to demand-planning workflows.
- Further production validation would be required before deployment.

2. Staffing and Store Operations

- Promotion-active days were associated with approximately 58.3% higher customer footfall.
- Average spend per customer was approximately 13.9% higher.
- Promotion calendars could therefore be considered when planning staffing and store capacity.

3. Store-Specific Planning

- Sales patterns differ substantially across stores and store types.
- Store-specific forecasting and inventory strategies could be considered rather than applying identical assumptions across all stores.

4. Promotion Planning

- Promotions were associated with higher sales across 1,114 of the 1,115 stores with both promotion and non-promotion observations.
 - This consistency suggests that promotion status is a useful predictive variable for demand forecasting.
 - The analysis does not establish that promotions caused the observed increase in sales.
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8. Overall Project Conclusion

The project evaluated retail demand using exploratory analysis, classical time series forecasting, machine learning, and statistical promotion analysis.

Key findings:

- Retail demand shows strong weekly and seasonal patterns.
- Seasonal Naive improved substantially over the basic Naive benchmark.
- SARIMA was the strongest classical forecasting model.
- LightGBM outperformed all evaluated forecasting models.
- LightGBM reduced MAE by 71.15% and RMSE by 70.71% compared with SARIMA.
- Lag, rolling, calendar, and promotion features contributed to the machine learning forecasting approach.
- Promotion-active days had 38.77% higher average sales than non-promotion days among open-store observations.
- 1,114 of 1,115 stores showed higher average sales during promotions.
- OLS estimated a +2,293.66 association after controlling for store and calendar factors.
- Because the data is observational, the promotion results should be interpreted as associations rather than causal effects.

The overall project demonstrates how exploratory analysis, statistical methods, classical forecasting, machine learning, and business interpretation can be combined to analyze retail demand and promotion patterns.